Documents Networks

The documents' network should serve to show the degree correlation between documents, and to define assortativity, which is degree homophily, and thus allows to show the polarization of users.

In the **documents_networks folder**, you can find two folders, each of them representing a community (designers, programmers). Each of these communities show:

- the degree distribution, which can show if the networks follow the Power Law distribution, e.g if $\gamma \in [2, 3]$;
- the heatmap showing degree correlation, which can show assortativity if it shows a pattern going from top left to bottom right, disassortativity otherwise.

These are using degrees, without considering upvotes' weights.

Semantic Networks

Semantic Networks should serve to show which words are used inside each community.

In the **semantic_networks folder**, you can find two folders, each of them representing a community (designers, programmers). Each of these communities contain two images, which are different types of visualization for the same network.
The nodes were filtered using degree (keeping around 500 words), while the importance of each node is defined through PageRank.

Topics

Topics should serve to identify what is being talked about inside a specific community through wordclouds.

In the **topics folder**, we have two folders that identify communities. For each of them, we have two folders that identify which model was used for those specific topics (BERTopic, Louvain). The last of these folders have various files:

- C, which identifies the clusters' matrix, linking a specific document ID to each topic;
- metrics, which show the values of NMI, modularity, Ncut and InfoMAP;
- a wordcloud, showing the top 100 words for each topic, with the most predominant ones being taken out if they were keywords or if they were present in the majority of other topics.

For the Louvain algorithm, it was executed 10 times for each community, trying to search for the best topics.

Emotions (Emotion Analysis)

Emotion analysis should serve to show how the emotions evolved over quarters from 2020 to 2025, where each value identifies the mean score for each emotion, showing how "powerful" that emotion is in a specific quarter. The emotions used are anger, fear, pessimism, joy, trust and optimism.

In the **emotions folder**, you can find the emotions' graphs for the whole period for each community (designers, programmers), as well as the emotions' graphs for both communities together in the same period.
In the **emotion/topics folder**, we have two folders that identify which model was used for those specific topics (BERTopic, Louvain), with each of them having both communities' topics emotions graphs.

Notice how the upvotes are not correlated to emotions, so the above-mentioned graphs use pure emotion scores obtained from the Hugging Face RoBERTa emotions model.

Sentiment (Sentiment Analysis)

Sentiment analysis should serve to show how the sentiment evolved over time, both quarterly and yearly.

In the **sentiment folder**, you can find three folders:

- **not_weighted**, containing the mean sentiment over quarters from 2020 to 2025, where each value identifies the mean sentiment score, both for all communities and for the two communities (designers, programmers) against each other. There is also an alternative visualization, defined as net_sentiment, which shows the same data of the two communities against each other using a line plot;
- **weighted**, containing the mean sentiment over quarters from 2020 to 2025, where each value identifies the weighted mean sentiment score, both for all communities and for the two communities (designers, programmers) against each other. There are also graphs showing the numbers of upvotes for each quarter, used to show the correlation. There is also an alternative visualization, defined as net_sentiment, which shows the same data of the two communities against each other using a line plot;
- **yearly,** containing the mean sentiment over each year, from 2020 to 2025, where each value identifies positive, neutral and negative scores for the sentiment. This is shown for all communities and for both of them in the same graph.

Notice how the weight is defined through the upvotes using the relation $s = \log(1 + |u|)$, since there is a negative correlation between upvotes and sentiment. The absolute value of the upvotes is used since in Reddit they can be both positive and negative.
The above-mentioned graphs use sentiment scores obtained from the Hugging Face RoBERTa sentiment model.